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BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DEEE – FIFTH SEMESTER EXAMINATION

ELECTRICAL UTILIZATION AND TRACTION

Time : 3 Hours] [*Total Marks* : 80

PART—A 3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define *(a)* Luminous flux, *(b)* Luminous intensity and *(c)* Lumen.

2. State any three requirements of good lighting.

3. State the advantages of electrical heating.

4. State the requirements of good heating element.

5. State the need of power saving devices.

6. List any six advantages of CFL lamps over INDICANDICENT LAMPS.

7. List various methods of track electrification.

8. List the types of services in electric traction.

9. State any six major equipment in traction sub-station.

10. Mention the requirements of railway coach air conditioning.

PART—B

8×5=40

Instructions : (1) Answer **all** questions.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* A light should has an output of 500 C.P in all directions below the lamp level. It is suspended at 3 mtr. Above the ground. Calculate

(i) Illumination directly below the lamp at the working plane.

(ii) Illumination at a point 2 mtr. away the horizontal plane from vertically below the lamp.

(iii) Total flux of light within a circular area of 1 mtr. dia. around the point vertically below the lamp.

(OR)

(b) Twp lamps of 500 watts each with an efficiency of 25 lumens/watt are mounted on two lamp posts 10 mtr. apart. The post are 3 mtr. and 5 mtr. height. Find the illumination

(i) just below the lamps

(ii) at a point mid way between the lamps

12. *(a)* Explain direct arc furnace with figures.

(OR)

(b) Explain different method of temperature control resistance furnaces with diagrams.

13. *(a)* *(i)* Write any four aspects of comparison of LED, CFL lamps an incandescent lamps.

(ii) State the requirements of train lighting.

(OR)

(b) *(i)* Explain the automatic temperature control with block diagram.

(ii) Explain MID-ON generation with a neat sketch.

14. *(a)* The speed-time curve of train consists of :

(i) Uniform acceleration of 6 kmphps for 25s.

(ii) Free running for 10 minutes

(iii) Uniform retardation of 6 kmphps to stop the train

(iv) A stop of 5 munites

Find the distance between the stations, the average and schedule speeds.

(OR)

(b) An electric train weighing 200 tonne has eight motors geared driving wheels, each wheel is 90 cm diameter. Determine the torque developed by each motor to accelerate the train to a speed of 48 kmph in 30 sec up-gradient of 1 in 200. The tractive resistance is 50 Newton's per tonne, the effect of rotational inertia is 10% of the train weight, the gear ratio is 4 to 1 and gearing efficiency is 80%.

15. *(a)* State the importance of the speed-time curve and explain them.

(OR)

(b) Derive the expression for total tractive effort for acceleration to overcome gravity pull and train resistance.

PART—C

10×1=10

Instructions : (1) Answer the following question.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. A drawing hall 30 m × 15 m with ceiling height of 5 m is to be provided with general illumination of 120 lux. Taking the coefficient of utilization of 0.5 and depreciation factor of 1.4, determine the no. of fluorescent tubes required, their spacing, mounting height and total wattage. Take luminous efficiency of fluorescent tubes 40 lumens/watt for 80 watt tube.

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